International Financial Statement Analysis

Chapter 1: Introduction to Financial Statement Analysis

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Learning Outcomes

- Describe the roles of financial reporting and financial statement analysis
- Describe the roles of the financial statements in evaluating a company's performance and financial position
- Describe the importance of financial statement notes and supplementary information
- Describe the objective of audits of financial statements
- Identify and describe information sources that analysts use in financial statement analysis
- Describe the steps in the financial statement analysis framework.

Introduction

- Financial analysis is the process of examining a company's performance in the context of its industry and economic environment in order to arrive at a decision or recommendation:
  - providing capital to companies—specifically, whether to invest in the company's debt or equity securities and at what price.
  - An investor in debt securities is concerned about the company's ability to pay interest and to repay the principal.
  - An investor in equity is concerned about the company's ability to pay dividends and the likelihood that its share price will increase.

Introduction

- Overall, a central focus of financial analysis is evaluating the company's ability to earn a return on its capital at least equal to the cost of that capital, to profitably grow its operations, and to generate sufficient cash to meet obligations and pursue opportunities.

Financial Reporting Vs. Financial Statement Analysis

- The role of financial statements issued by companies is to provide information about a company's performance, financial position, and changes in financial position that is useful to a wide range of users in making economic decisions.

- The role of financial statement analysis is to use financial reports prepared by companies, combined with other information, to evaluate the past, current, and potential performance and financial position of a company for the purpose of making investment, credit, and other economic decisions.

Economic decisions

- Evaluating an equity investment for inclusion in a portfolio.
- Evaluating a merger or acquisition candidate.
- Evaluating a subsidiary or operating division of a parent company.
- Deciding whether to make a venture capital or other private equity investment.
- Determining the creditworthiness of a company in order to decide whether to extend a loan to the company and if so, what terms to offer.

Economic decisions (cont’)

- Extending credit to a customer.
- Examining compliance with debt covenants or other contractual arrangements.
- Assigning a debt rating to a company or bond issue.
- Valuing a security for making an investment recommendation to others.
- Forecasting future net income and cash flow.

Primary Financial Statements and Other Information Sources

- Information from outside the company will likely include economic statistics, industry reports, trade publications, and databases containing information on competitors.
- The company itself provides core information for analysis in its financial reports, press releases, investor conference calls, and webcasts.
- Companies prepare financial reports at regular intervals (annually, semiannually, and/or quarterly, depending on the applicable regulatory requirements)
- Financial statements are the result of an accounting process that records a company's economic activities, following the applicable accounting standards and principles.
- Financial statements are almost always audited by independent accountants

Financial Statements and Supplementary Information

- A complete set of financial statements includes:
  - A statement of financial position (i.e., a balance sheet),
  - A statement of comprehensive income (i.e., a single statement of comprehensive income or an income statement and a statement of comprehensive income),
  - A statement of changes in equity, and
  - A statement of cash flows.

Additional information

- A letter from the chairman of the company,
- A report from management discussing the results (typically called management discussion and analysis (MD&A) or management commentary),
- An external auditor's report providing assurances,
- A governance report describing the structure of the company's board of directors, and a corporate responsibility report.

Balance Sheet

€ million	Dec. 31, 2017	Dec. 31, 2016
Assets		
Noncurrent assets		
Intangible assets	63,419	62,599
Property, plant, and equipment	55,243	54,033
Lease assets	39,254	38,439
Investment property	468	512
Equity-accounted investments	8,205	8,616
Other equity investments	1,318	996
Financial services receivables	73,249	68,402
Other financial assets	8,455	8,256
Other receivables	2,252	2,009
Tax receivables	407	392
Deferred tax assets	9,810	9,756
	262,081	**254,010**
Current assets		
Inventories	40,415	38,978
Trade receivables	13,357	12,187
Financial services receivables	53,145	49,673
Other financial assets	11,998	11,844
Other receivables	5,346	5,130
Tax receivables	1,339	1,126
Marketable securities	15,939	17,520
Cash, cash equivalents, and time deposits	18,457	19,265
Assets held for sale	115	–
	160,112	**155,722**
Total assets	**422,193**	**409,732**

Equity and Liabilities		
Equity		
Subscribed capital	1,283	1,283
Capital reserves	14,551	14,551
Retained earnings	81,367	70,446
Other reserves	560	(1,158)
Equity attributable to Volkswagen AG hybrid capital investors	11,088	7,567
Equity attributable to Volkswagen AG shareholders and hybrid capital investors	108,849	92,689
Noncontrolling interests	229	221
	109,077	**92,910**
Noncurrent liabilities		
Financial liabilities	81,628	66,358
Other financial liabilities	2,665	4,488
Other liabilities	6,199	5,664
Deferred tax liabilities	5,636	4,745
Provisions for pensions	32,730	33,012
Provisions for taxes	3,030	3,556
Other provisions	20,839	21,482
	152,726	**139,306**
Current liabilities		
Put options and compensation rights granted to noncontrolling interest shareholders	3,795	3,849
Financial liabilities	81,844	88,461
Trade payables	23,046	22,794
Tax payables	430	500
Other financial liabilities	8,570	9,438
Other liabilities	15,961	15,461
Provisions for taxes	1,397	1,301
Other provisions	25,347	35,711
	160,389	**177,515**
Total equity and liabilities	**422,193**	**409,732**

The balance sheet of the Volkswagen Group in accordance with International Financial Reporting Standards (Excerpt).

Balance Sheet (cont')

	As of January 31	
(Amounts in $ millions)	**2018**	**2017**
ASSETS		
Current assets:		
Cash and cash equivalents	6,756	6,867
Receivables, net	5,614	5,835
Inventories	43,783	43,046
Prepaid expenses and other	3,511	1,941
Total current assets	59,664	57,689
Property and equipment:		
Property and equipment	185,154	179,492
Less accumulated depreciation	(77,479)	(71,782)
Property and equipment, net	107,675	107,710
Property under capital lease and financing obligations:		
Property under capital lease and financing obligations	12,703	11,637
Less accumulated amortization	(5,560)	(5,169)
Property under capital lease and financing obligations, net	7,143	6,468
Goodwill	18,242	17,037
Other assets and deferred charges	11,798	9,921
Total assets	204,522	198,825
LIABILITIES AND EQUITY		
Current liabilities:		
Short-term borrowings	5,257	1,099
Accounts payable	46,092	41,433
Accrued liabilities	22,122	20,654
Accrued income taxes	645	921
Long-term debt due within one year	3,738	2,256
Capital lease and financing obligations due within one year	667	565
Total current liabilities	78,521	66,928
Long-term debt	30,045	36,015
Long-term capital lease and financing obligations	6,780	6,003
Deferred income taxes and other	8,354	9,344
Commitments and contingencies		
Equity:		
Common stock	295	305
Capital in excess of par value	2,648	2,371
Retained earnings	85,107	89,354
Accumulated other comprehensive loss	(10,181)	(14,232)
Total Walmart shareholders' equity	77,869	77,798
Noncontrolling interest	2,953	2,737
Total equity	80,822	80,535
Total liabilities and equity	204,522	198,825

Walmart Consolidated Balance Sheet (Excerpt).

Statement of Comprehensive Income

€ million	2017	2016
Sales revenue	**230,682**	**217,267**
Cost of sales	(188,140)	(176,270)
Gross result	**42,542**	**40,997**
Distribution expenses	(22,710)	(22,700)
Administrative expenses	(8,254)	(7,336)
Other operating income	14,500	13,049
Other operating expenses	(12,259)	(16,907)
Operating result	**13,818**	**7,103**
Share of the result of equity-accounted investments	3,482	3,497
Interest income	951	1,285
Interest expenses	(2,317)	(2,955)
Other financial result	(2,022)	(1,638)
Financial result	**94**	**189**
Earnings before tax	**13,913**	**7,292**
Income tax income/expense	(2,275)	(1,912)
Current	(3,205)	(3,273)
Deferred	930	1,361
Earnings after tax	11,638	5,379
of which attributable to		
Noncontrolling interests	10	10
Volkswagen AG hybrid capital investors	274	225
Volkswagen AG shareholders	11,354	5,144
Basic earnings per ordinary share in €	**22.63**	**10.24**
Diluted earnings per ordinary share in €	**22.63**	**10.24**
Basic earnings per preferred share in €	**22.69**	**10.30**
Diluted earnings per preferred share in €	**22.69**	**10.30**

The income statement of the Volkswagen Group (Excerpt).

Statement of Comprehensive Income (cont')

€ million	Fiscal Year Ended December 31, 2017
Earnings after tax	**11,638**
Pension plan remeasurements recognized in other comprehensive income	
Pension plan remeasurements recognized in other comprehensive income, before tax	785
Deferred taxes relating to pension plan remeasurements recognized in other comprehensive income	(198)
Pension plan remeasurements recognized in other comprehensive income, net of tax	588
Share of other comprehensive income of equity-accounted investments that will not be reclassified to profit or loss, net of tax	96
Items that will not be reclassified to profit or loss	**683**
Exchange differences on translating foreign operations	
Unrealized currency translation gains/losses	(2,095)
Transferred to profit or loss	(4)
Exchange differences on translating foreign operations, before tax	(2,099)
Deferred taxes relating to exchange differences on translating foreign operations	(8)
Exchange differences on translating foreign operations, net of tax	(2,107)
Cash flow hedges	
Fair value changes recognized in other comprehensive income	6,137
Transferred to profit or loss	(558)
Cash flow hedges, before tax	5,579
Deferred taxes relating to cash flow hedges	(1,597)
Cash flow hedges, net of tax	3,982
Available-for-sale financial assets	
Fair value changes recognized in other comprehensive income	56
Transferred to profit or loss	62
Available-for-sale financial assets, before tax	118
Deferred taxes relating to available-for-sale financial assets	(25)
Available-for-sale financial assets, net of tax	93
Share of other comprehensive income of equity-accounted investments that may be reclassified subsequently to profit or loss, net of tax	(346)
Items that may be reclassified subsequently to profit or loss	**1,622**
Other comprehensive income, before tax	4,133
Deferred taxes relating to other comprehensive income	(1,828)
Other comprehensive income, net of tax	**2,305**
Total comprehensive income	**13,943**

The Statement of Comprehensive Income of the Volkswagen Group (Excerpt)

Statement of Changes in Equity

€ million	Subscribed capital	Capital reserves	Retained earnings	Total equity
Balance at 1 January 2017	1,283	14,551	70,446	92,910
Earnings after tax	–	–	11,354	11,638
Other comprehensive income, net of tax	–	–	586	2,305
Total comprehensive income	–	–	11,940	13,943
Capital increases	–	–	–	3,481
Dividends payment	–	–	–1,015	–1,332
Capital transactions involving a change in ownership interest	–	–	–	–
Other changes	–	–	–4	75
Balance at December 31, 2017	1,283	14,551	81,367	109,077

The Statement of Comprehensive Income of the Volkswagen Group (Excerpt, several interim columns were excluded from the presentation)

Cash Flow Statement

€ million	2017	2016
Cash and cash equivalents at beginning of period	**18,833**	**20,462**
Earnings before tax	13,913	7,292
Income taxes paid	(3,664)	(3,315)
Depreciation and amortization of, and impairment losses on, intangible assets, property, plant and equipment, and investment property	10,562	10,100
Amortization of and impairment losses on capitalized development costs	3,734	3,586
Impairment losses on equity investments	136	130
Depreciation of and impairment losses on lease assets	7,734	7,107
Gain/loss on disposal of noncurrent assets and equity investments	(25)	(222)
Share of the result of equity-accounted investments	274	377
Other noncash expense/income	(480)	716
Change in inventories	(4,198)	(3,637)
Change in receivables (excluding financial services)	(1,660)	(2,155)
Change in liabilities (excluding financial liabilities)	5,302	5,048
Change in provisions	(9,443)	5,966
Change in lease assets	(11,478)	(12,074)
Change in financial services receivables	(11,891)	(9,490)
Cash flows from operating activities	**(1,185)**	**9,430**
Investments in intangible assets (excluding development costs), property, plant and equipment, and investment property	(13,052)	(13,152)
Additions to capitalized development costs	(5,260)	(5,750)
Acquisition of subsidiaries	(277)	(119)
Acquisition of other equity investments	(561)	(309)
Disposal of subsidiaries	496	(7)

Cash Flow Statement (cont’)

Disposal of other equity investments	24	2,190
Proceeds from disposal of intangible assets, property, plant and equipment, and investment property	411	351
Change in investments in securities	1,376	(1,245)
Change in loans and time deposits	335	(2,638)
Cash flows from investing activities	**(16,508)**	**(20,679)**
Capital contributions	3,473	–
Dividends paid	(1,332)	(364)
Capital transactions with noncontrolling interest shareholders	–	(3)
Proceeds from issuance of bonds	30,279	14,262
Repayments of bonds	(17,877)	(23,601)
Changes in other financial liabilities	3,109	19,455
Lease payments	(28)	(36)
Cash flows from financing activities	**17,625**	**9,712**
Effect of exchange rate changes on cash and cash equivalents	(727)	(91)
Net change in cash and cash equivalents	**(796)**	**(1,628)**
Cash and cash equivalents at end of period	**18,038**	**18,833**

The Cash Flow Statement of the Volkswagen Group

Financial Notes and Supplementary Schedules

- A company's significant accounting choices (policies, methods, and estimates) must be discussed in the notes to the financial statements.
- Volkswagen's 2017 financial statements include more than 100 pages of notes. For example:
  - Volkswagen discloses that its fiscal year corresponds to the calendar year;
  - Its financial statements are prepared in accordance with IFRS as adopted by the European Union;
  - The statements are prepared in compliance with German law;
  - The statements are denominated in millions of euros unless otherwise specified, and the figures have been rounded, which may result in minor discrepancies when added.
  - Volkswagen also states that its financial statements are on a consolidated basis—that is, including Volkswagen AG and all of the subsidiary companies it controls.

Management Commentary or Management's Discussion and Analysis

- Publicly held companies typically include a section in their annual reports where management discusses a variety of issues, including the nature of the business, past results, and future outlook.
- IFRS suggests that a "decision-useful management commentary" should contain:

1) the nature of the business;
2) management's objectives and strategies;
3) the company's significant resources, risks, and relationships;
4) results of operations; and
5) critical performance measures.

Auditor’s Reports

- International standards on auditing (ISAs) have been developed by the International Auditing and Assurance Standards Board (IAASB).
- Audits are designed and conducted using sampling techniques; therefore, auditors cannot express an opinion that provides absolute assurance about the accuracy or precision of the financial statements.
- Instead, the independent audit report provides reasonable assurance that the financial statements are fairly presented, meaning there is a high probability that the audited financial statements are free from material errors, fraud, or illegal acts that directly affect them.

Auditor's Reports (cont')

- An unqualified audit opinion: the financial statements give a "true and fair view" (international) or are "fairly presented" (international and US) in accordance with applicable accounting standards. (unmodified or clean)
- A qualified audit opinion: there is a scope limitation or an exception to accounting standards.
- An adverse audit opinion: the financial statements materially depart from accounting standards and are not fairly presented.
- A disclaimer of opinion: for some reason, such as a scope limitation, the auditors are unable to issue an opinion.

Other Sources of Information

- Proxy statements: distributed to shareholders about matters that are to be put to a vote at the annual (or special) meeting of shareholders (e.g. management and director compensation)
- Interim reports: provided semiannually or quarterly, depending on the applicable regulatory requirements (unaudited).
- Company's website, press releases, conference calls

Financial Statement Analysis Framework

Phase	Sources of Information	Output
1. Articulate the purpose and context of the analysis.	• The nature of the analyst's function, such as evaluating an equity or debt investment or issuing a credit rating. • Communication with client or supervisor on needs and concerns. • Institutional guidelines related to developing specific work product.	• Statement of the purpose or objective of analysis. • A list (written or unwritten) of specific questions to be answered by the analysis. • Nature and content of report to be provided. • Timetable and budgeted resources for completion.
2. Collect input data.	• Financial statements, other financial data, questionnaires, and industry/economic data. • Discussions with management, suppliers, customers, and competitors. • Company site visits (e.g., to production facilities or retail stores).	• Organized financial statements. • Financial data tables. • Completed questionnaires, if applicable.
3. Process data.	• Data from the previous phase.	• Adjusted financial statements. • Common-size statements. • Ratios and graphs. • Forecasts.
4. Analyze/interpret the processed data.	• Input data as well as processed data.	• Analytical results.
5. Develop and communicate conclusions and recommendations (e.g., with an analysis report).	• Analytical results and previous reports. • Institutional guidelines for published reports.	• Analytical report answering questions posed in Phase 1. • Recommendation regarding the purpose of the analysis, such as whether to make an investment or grant credit.
6. Follow-up.	• Information gathered by periodically repeating above steps as necessary to determine whether changes to holdings or recommendations are necessary.	• Updated reports and recommendations.